

AI based Training Manager for AOI? It's already here.

Training Manager empowers your team to retrain inspection models in real time—making quality control faster, smarter, and easier to manage.

The adoption of AI-driven optical inspection systems facilitates adaptability

The manufacturing world is evolving fast. Today's production environments - like electronics, automotive, and medical devices - demand systems that can not only spot anomalies with precision, but also adapt to constant change. This is where deep learning comes into play. Convolutional neural networks (CNNs), trained on visual datasets, detect subtle defects that traditional methods often miss. Beyond improving first-pass yield, these AI-powered systems reduce false positives, limit unnecessary interventions, and bring real-time intelligence to inspection lines. But what makes this transformation possible isn't just the AI, it's the ability to continuously train and evolve these models without starting from scratch each time.

AI and AOI? Faster model development with better results

Training a neural network used to be a job for data scientists. High-performing AOI systems relied on curated, pre-annotated datasets and long training cycles, making adaptation slow and expensive. But those days are over. Modern AI-based inspection platforms now offer intuitive tools to annotate and train models quicker. Operators can teach systems to recognize new defects or components without disrupting production or waiting for centralized programming teams.

Economic and operational benefits of intelligent inspection

Research shows that manufacturers who have adopted AI-native AOI and SPI systems report clear returns: up to 30% faster inspection cycles and a 25% reduction in product returns [1][2]. When AI takes on the heavy lifting—sorting normal from anomalous patterns, your teams are freed up for more strategic tasks, whether that's process improvement, analytics, or innovation. Still, many current solutions struggle with retraining. Updating an inspection model often means wiping out prior knowledge — what's known as catastrophic forgetting. And most vendors still require specialized technicians to preprocess and annotate images

Training Manager: Correct, organize, and train—without interrupting production

That's where Delvitech's Training Manager stands out. Built directly into the company's Neith software platform, this tool gives

manufacturers a way to annotate, correct, and retrain inspection models with just a few clicks. Through an intuitive web-based interface, operators can:

- Upload and label images
 - Correct misclassifications
 - Organize annotated datasets
 - Train and manage multiple neural network models
 - Monitor training history for traceability
- And it works fast: training sessions take under a minute and can run on offline stations—keeping live lines fully operational. Training Manager currently supports recognition for:
- Component bodies
 - Pins
 - Solder joint quality

It's a comprehensive environment where both classification and semantic segmentation are supported. Users can draw detailed polygons to define specific features, like a screw head or a solder paste deposit, then train a model to identify them with high accuracy. Even just a single image is enough to trigger a meaningful retraining cycle, letting your team correct edge cases and rare defects on the fly.

Real-world results: practical and measurable

How does this play out on the shop floor?

Body segmentation: When the system misses a non-standard component like a screw, the operator can label it using the Annotation Tool. After training, the network not only detects the screw but also analyzes details like proper fastening.

Pin Segmentation: Initially missed, a pin is manually annotated with the Annotation tool and added to training. The model now recognizes it across the board.

PTH Resistor Segmentation: The system initially fails to detect a PTH resistor, not having been trained on this component type. Using the Annotation tool, the operator marks it, retrains the model, and the system then reliably recognizes it every time onwards.

Paste deposit detection: The system initially misses the solder paste deposits, lacking prior training on them. With the Annotation tool, the operator labels just a few deposits and starts training. The network then accurately and autonomously detects all deposits.

Wire bonding: Same as the paste deposit detection, with a few quick labels, the system learns to detect previously unseen wire bonds in under a minute.

Pin in Paste Segmentation: By labeling a few key examples with the Annotation tool, the



operator starts the training process over pin-in-paste scenario. Afterward, the network accurately identifies all features.

Solder joint correction: When lighting conditions lead to a misclassification of a solder joint, the operator can manually reclassify it as good, and this correction is absorbed by the neural network.

Learning without forgetting: Knowledge Retention Technology (KRT)

One of the biggest differentiators in Delvitech's approach is its proprietary Knowledge Retention Technology (KRT). Unlike conventional systems that overwrite previous training during updates, KRT allows models to learn continuously—adding new information without erasing existing knowledge. Thanks to KRT, every new training iteration enhances the model's capabilities—without resetting progress.

Delvitech and its Training Manager

While many in the industry are still talking about AI integration, Delvitech's AI-native approach is already delivering results—with a tool designed specifically to empower your team: the Training Manager.

With it, manufacturers can achieve:

- **Shorter model adaptation times**
- **Higher inspection accuracy and speed**
- **Consistent quality across multiple production environments**

In a competitive landscape defined by speed, complexity, and zero-defect expectations, Training Manager offers a practical, scalable solution that puts your team, not the algorithm, in control of inspection performance. AI-based training isn't coming. **It's already here.**

[1]Intellic (2025). Quality Inspection Trends in Manufacturing Using AI in 2025.

<https://intellic.com/quality-inspection-trends-in-manufacturing-using-ai-in-2025>

[2]Data Bridge Market Research. Global Automated Optical Inspection Market Report.

<https://www.databridgemarketresearch.com/reports/global-automated-optical-inspection-market>